OOP

Q. 1 Write a program to accept marks of three subject and display total and percentage.

→
```
#include <iostream.h>
#include <conio.h>

void main()
{
    int, m1, m2, m3, total;
    float per;

    cout<<" Enter the marks of subject 1:";
    cin>>m1;
    cout<<" Enter the marks of subject 2:";
    cin>> m2;
    cout<<" Enter the marks of subject 3:";
    cin>> m3;

    total = m1 + m2 + m3;
    per = float (total/3);

    cout<<" total is : " << total << endl;
    cout<<" Percentage is : " << per << endl;

    getch();
}
```

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Q. 2 Write a program to accept percentage of student and display its grade using nested if else

→

```
# include <iostream.h>
# include <conio.h>

void main()
{
    float percentage;
    clrscr();

    cout << " Enter the percentage : ";
    cin >> percentage;

    if ( percentage <= 100)
    {
      if ( percentage >= 75)
      {
        cout << " A Grade ";
      }
      else if (percentage >= 60 )
      {
        cout << " B grade ";
      }
      else if (percentage >= 50)
      {
        cout << " C grade ";
      }
      else if (percentage >= 40)
      {
        cout << " D grade ";
      }
```

```
else
{
  cout<<"fail";
 }
}
 else
 {
    cout<<" Invalid input... Enter valid
            percentage:";
 }
  getch();
}
```

Q 3 Write a program to accept numbers using array and display smallest among it.

→

```
# include <iostream.h>
# include < conio.h >

void main()
{
        int arr[5];
        int i, min=0;
        clrscr();

        cout<<" Enter the 5 elements:";
        for( i=0 ; i<5 ; i++)
        {
            cin>> arr[i];
        }
        for (i=0 ; i<5 ; i++)
        {
            if (arr[i]< min)
```

```
{
    min = arr[i]
  }
  }
  cout<<" Smallest number = "<<min;
  getch();
}
```

Q 4 Write a program to accept numbers using array and display its addition.

→

```
#include<iostream.h>
#include<conio.h>

void main()
{
    int arr[5];
    int i, sum=0;
    clrscr();

    cout<<" Enter the 5 elements:";
    for(i=0; i<5; i++)
    {
        cin>>arr[i];
    }
    for(i=0; i<5; i++)
    {
        sum+= arr[i];
    }
    cout<<" sum = "<<sum;
    getch();
}
```

Q.5 Write a program to create a structure of a book having members id, name, price, perform accept & display operation for 2 books.

→
```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>

Struct book
{
    int id;
    char name [50];
    int price;
} b[2];

void main()
{
    int i;
    clrscr();

    for (i=0; i<2; i++)
    {
        cout<<"Enter the id of book:";
        cin>> b[i].id;
        cout<<"Enter the name of book:";
        cin>> b[i].name;
        cout<<"Enter the price of book:";
        cin>> b[i].price;
    }
```

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```
for(i=0; i<2; i++)
{
    cout<<"Details of book" <<i+1;
    cout<<"ID of book: " << b[i].id;
    cout<<" Name of book:" << b[i].name;
    cout<<" Price of book:" << b[i].price;
}
getch();
}
```

Q. 6. Write a program to create a class book having members id, name and price. Perform accept and display operation for 2 books using object.

→

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>

class book
{
    public:
    char name[50];
    int id;
    int price;

    void accept()
    {
        cout<<" Enter the id of book:";
        cin>>id;
        cout<<"Enter the name of book:";
        gets(name);
```

```
    cout << " Enter the price of book:";
    cin>> price;
  }
  void display()
  {
    cout<<" Book Details:";
    cout<<" ID: "<< id<<endl;
    cout<<" Name: "<< name<<endl;
    cout<<" Price: "<< price<<endl;
  }
};
void main()
{
    clrscr();
    book b1, b2;
    b1.accept();
    b1.display();
    b2.accept();
    b2.display();
    getch();
}
```

Q.7 Write a program to create a class book having members id, name and price. perform accept and display operation for 5 books using array of object.

→

```
#include <iostream.h>
#include <conio.h>
#include<stdio.h>

class book
{
```

```
public:
int ~~id~~ b_id;
char b_name[20];
Float price;

void getdata( int b_id, char bname[20],
              float p)
{
    b_id = id;
    b_name = bname;
    price = p;
}
void display()
{
    cout<<" ID:  "<<b_id<<endl;
    cout<<" Name:  "<<bname<<endl;
    cout<<" price:  "<<price<<endl;
}
};
int main()
{
    clrscr();
    book b[5];
    int id, i;
    char bname[20];
    Float p;
    for (i=0; i<5; i++)
{
    cout<<endl<<" Enter details of"<<i+1<<" the
    book"<<endl;
    cout<<" Enter book id: ";
    cin>>id;
```

```
cout<<" Enter book name :";
gets(bname);
cout<<" Enter book price :";
cin>>p;
b[i].getdata();
}

for(i=0;i<5;i++)
{
cout<< endl <<" Details of "<<i+1<<" the book:";
b[i].display();
}

getch();
}
```